HANNING LU

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EDUCATION

University of Leeds Sep 2024 – Jun 2027 (Expected)
Bachelor of Science in Computer Science | Leeds, United Kingdom | GPA:3.5/4.0

RESEARCH EXPERIENCE

Compiler Optimization and LLVM IR Decompilation May 2025 - Sep 2026
Prof. Zheng Wang & Dr. Chunwei Xia ,University of Leeds | Research Assistant

- Built a **closed-loop compiler agent** that plans, executes, profiles, and revises optimization actions from runtime and compiler feedback.
- Automated LLVM pass selection, parameter modification, and configuration tuning with correctness-gated iterative execution.
- Achieved **2.25× geomean speedup over -O3** on PolyBench through autonomous compiler optimization. (first author; preprint, 2026)
- Raised LLVM IR decompilation accuracy from **~50% to 90%+** through iterative LLM self-correction. (First-author paper, ACM ICS 2026 Workshop)
- Revealed that **higher revision propensity does not imply higher revision reliability** in LLM cascades.(Co-first author; preprint, 2026)

Efficient LLM Inference and Model Optimization Jun 2024 - Jun 2025
Prof. Tong Yang's Group, Peking University | Research Assistant

- Achieved 70% KV-cache compression with only ~1% performance loss on Llama-2-7B/LongBench using adaptive token retention across heads and decoding stages.
- Explored quantization, structured/unstructured pruning, and sparse-weight inference for lower-memory, lower-latency LLM deployment. (Co-author, arXiv:2409.11650)
- Built a context-aware retrieval and controlled-generation pipeline that dynamically routes relevant external content into LLM outputs.

Graph Representation Learning & LLM Post-Training for Trustworthy AI Oct 2025 - Jul 2026
Bin Chong's Group, Peking University | Research Assistant

- Developed **adaptive hyperbolic graph learning** to model hierarchical and heterogeneous structures, improving Weibo accuracy from 89.0% to **91.5%.** (First author; submitted to NeurIPS 2026)
- **Post-trained Qwen3-0.6B** with **LoRA-SFT** and **Active-GRPO**, using gated grounding and evidence-quality rewards for reliable reasoning.
- Achieved **98.9% / 98.8%** accuracy on Fox8-23 / BotSim-24 with near-100% evidence coverage and fully grounded decisions.
- Built an **auditable decision framework** combining graph models with an independently trained semantic verifier, supporting evidence tracing and conflict-aware abstention. (First author; submitted to AAAI 2027).

TPS-Bench:Multi-Tool Agent Planning & Scheduling Jun 2025 - Aug 2025
Prof. Zhijie Deng's Group, Shanghai Jiao Tong University | Research Assistant

- Contributed to **TPS-Bench**, an MCP-based benchmark and agent framework for task decomposition, tool selection, and coordinated multi-step execution.
- Improved **multi-tool planning reliability and fault recovery** through adaptive tool reselection, execution scheduling, and intermediate-result aggregation. (code released on GitHub)

Knowledge-Graph Reasoning and Real-Time Voice AI Jun 2025 - Aug 2025
Prof. Jun Wei's Group, Tsinghua University | Research Assistant

- Built **multi-hop knowledge-graph reasoning** with structured graph traversal and cross-node evidence aggregation for complex question answering.
- Enabled **low-latency real-time voice interaction** by integrating LLM inference, incremental generation, streaming TTS, and audio delivery into an end-to-end serving pipeline.

Hierarchical Generative Modeling for Single-Cell Data Sep 2025 - Feb 2026

Prof. Ziqing Li's Group, Westlake University | Research Assistant

- Achieved **best overall fate-tree reconstruction**, reaching 0.966 Core4 at 42k cells while maintaining stable scaling.
- Recovered ground-truth lineage structure with ($R>0.95$) temporal fidelity on C. elegans.
- Outperformed generative baselines with **93.5% diversity and 10.4 FID**. (manuscript in preparation)

Loop Transformers and Recurrent-Compute Reasoning May 2026 - Sep 2026

Prof. Menglin Yang's Group, HKUST (Guangzhou) | Research Assistant

- Revealed **high local redundancy and rapid output saturation** in frozen recurrent LLMs, explaining why deeper loops rarely improve performance.
- Verified across **diverse tasks and extensive experimental settings** that frozen loops yield no stable gains beyond noise.
- Identified **learned task-aligned updates** as the key to effective recurrent computation.

SELECTED PUBLICATIONS

- **H. Lu**, C. Xia. *Agent-Agnostic End-to-End C/C++ Application Performance Optimization*. Proceedings of the 40th ACM International Conference on Supercomputing Workshops, 2026. DOI: 10.1145/3774895.3812199.
- **H. Lu**, Y. Yang, J. Su, Y. Liu, Z. Yao, Y. Li, T. Liang, Z. Zhang, R. Ran, K. Xu, B. Chong. *SAHG: Sector-Anisotropic Hyperbolic Graph Model for Social Bot Detection*. In submission to NeurIPS 2026; arXiv preprint, 2026. arXiv:2605.30166.
- **H. Lu**, K. Hu, Y. Yang, Y. Liu, Z. Yao, Y. Li, Y. Chen, C. Zhang, H. Zheng, K. Xu, C. Ran, B. Chong. *VeriBot: Evidence-Routed Social Bot Detection with Auditable Decision Chains*. In submission to AAAI 2027.
- K. Hu, Y. Yang, **H. Lu**, Y. Liu, Y. Chen, C. Zhang, H. Zheng, K. Xu, C. Ran, H. Peng, P. S. Yu, B. Chong. *BotRoute: Dual-Branch Selective Routing for Social Bot Detection*. In submission to KDD 2027.
- Z. Yao, Y. Tang, **H. Lu**, Y. Li, M. Yang. *Hierarchy-Aware Sparse Autoencoders via Activation-Cone Routing*. In submission to AAAI 2027.
- Z. Yao, **H. Lu**. *HCGM: Hyperbolic Code Graph Modeling for Repository-Level Software Intelligence*. In submission to KDD 2027.
- H. Jia, Y. Liu, B. Chong, Y. Yang, Y. Chen, J. Liang, Q. Li, **H. Lu**, K. Xu, H. Zheng, C. Zhang, H. Peng, P. S. Yu. *FinHarness: An Inline Lifecycle Safety Harness for Finance LLM Agents*. arXiv preprint, 2026. arXiv:2605.27333.
- Y. Wang, T. Yang, X. Liang, G. Wang, **H. Lu**, Z. Xu, Y. Li, W. Li. *Art and Science of Quantizing Large-Scale Models: A Comprehensive Overview*. arXiv preprint, 2024. arXiv:2409.11650.
- W. Zhong*, **H. Lu***, Z. Yao, S. Zhang, C. Xia. *Revision Propensity Is Not Revision Reliability: Ground-Truth-Conditioned Evaluation of Prompt Framing in LLM Cascades*. Preprint, 2026. **Co-first authors*.

COMPETITIONS & HONORS

- Gold Medal, CCPC Zhengzhou Invitational - Team Captain 2024
- Silver Medal, 48th ICPC Xi'an Invitational - Team Captain

INTERNSHIP EXPERIENCE

Artificial Intelligence Engineer Intern Jun 2025 - Sep 2025

Sheet0 (AI Startup)

- Designed and optimized multi-agent coordination code to improve data-collection accuracy and runtime stability; built configurable task-scheduling and execution workflows that increased fault tolerance and recovery in complex scenarios.

Video Generation Infrastructure Engineer Intern Jun 2026 - Sep 2026

Xingjie Intelligence (AI Startup)

- Develop SGLang-based serving infrastructure for Wan-series diffusion models using tensor, pipeline, and sequence parallelism, KV caching, and block-wise autoregressive generation.
- Profile GPU bottlenecks with Nsight Systems, Nsight Compute, and PyTorch Profiler; evaluate FP8, weight-only, and activation quantization with CUDA Graph, kernel fusion, continuous batching, and streaming inference across **FPS, chunk latency, TTFT, memory use, and scaling efficiency**.